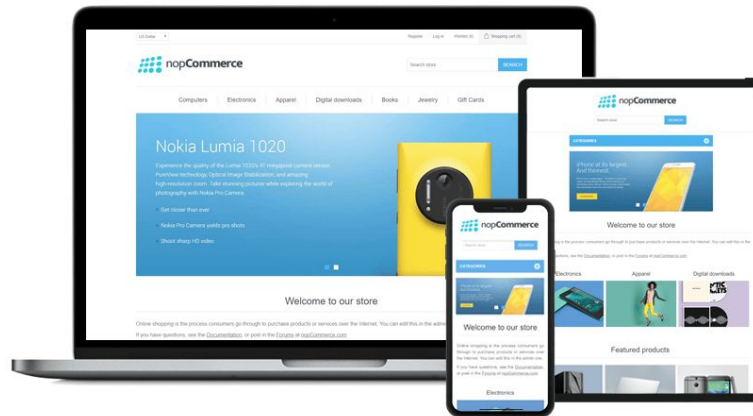




Architectural Evolution of nopCommerce

David Pelicano ◉ Henrique Ferreira ◉ João Monteiro ◉ Jorge Domingues

05.05.2026



Index

A Brief Context

B Evolved Architecture

C Final Architecture Decisions

D Demonstration Setup

E Demo

F Evidence Pack

G Known Limitations

H Conclusion

VerdeMart: Commerce Core



Initial State Analysis

In-process events

No delivery guarantee, no retry and no persistence: events vanish if a handler fails.

No inventory lock

Concurrent orders can oversell the same unit, leading to fulfilment conflicts.

No integration pattern

Every external connection is built from scratch

Quality Attributes Scenarios

QAS-1: Reliability

Zero orders lost during OpenBoxes outage

QAS-2: Consistency

Zero oversell under concurrent web + POS load

QAS-3: Availability

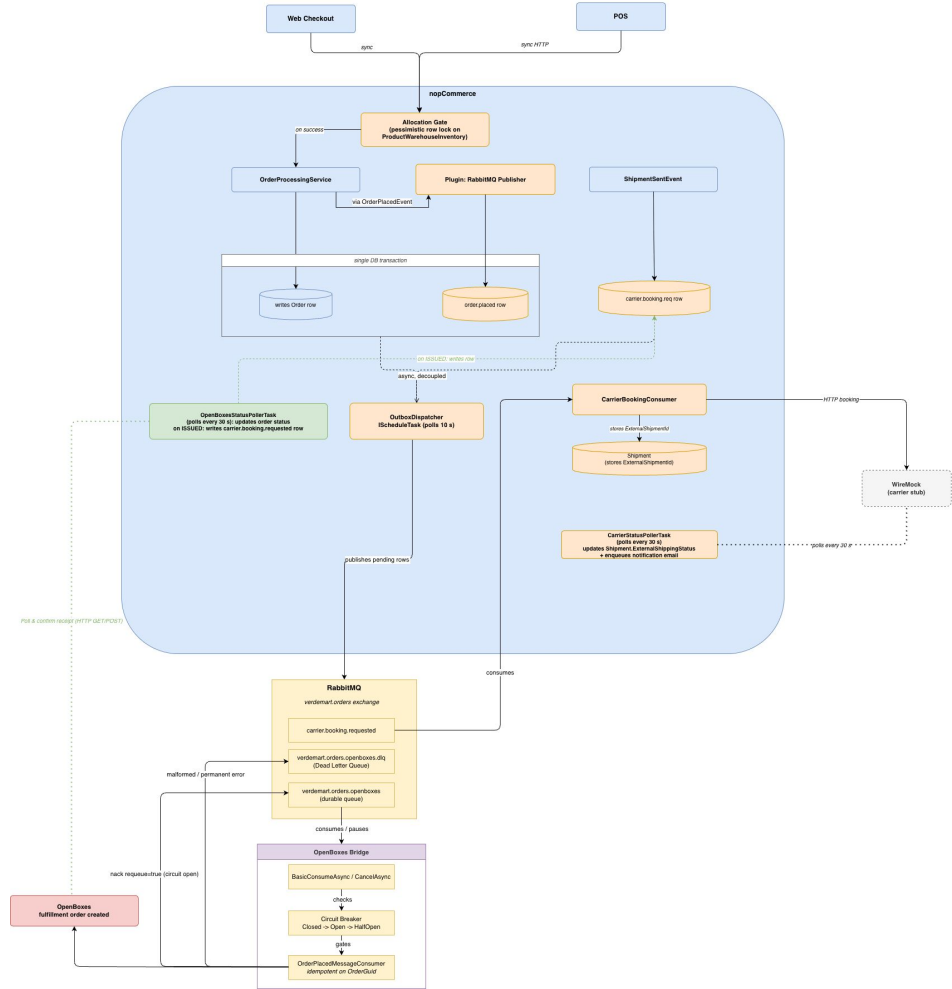
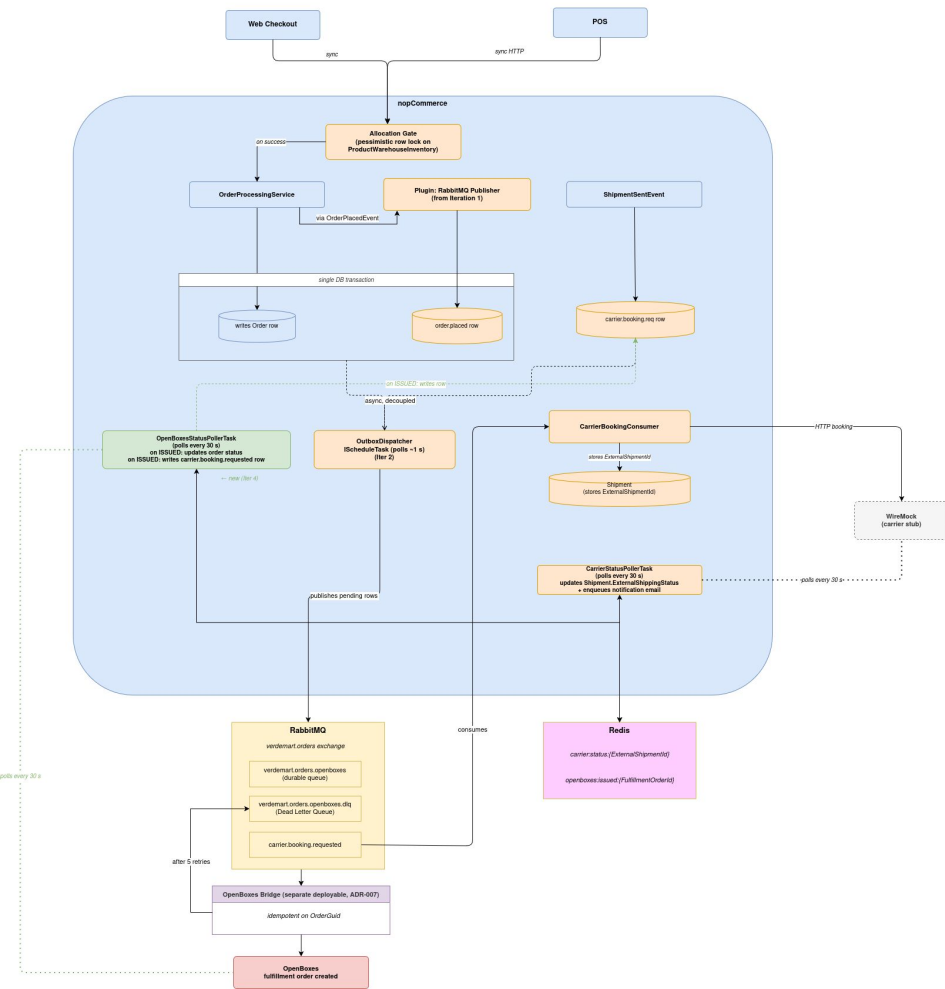
Checkout under 3s regardless of surrounding system latency

QAS-4: Recoverability

Missed events must self-heal after outage

QAS-5: Visibility

External state changes must reach the customer quickly



System Architecture



nopCommerce Core

MODIFIED

- **IProductService.cs**
interface hook for AllocationGate
- **OrderProcessingService.cs**
routed through AllocationGate at checkout



Standalone Service

BUILT

- **VerdeMart.OpenBoxesBridge**
`verdemart.orders.openboxes`
creates fulfillment orders in OpenBoxes
circuit breaker + dedup + DLQ



Plugins

BUILT FROM SCRATCH

- **Nop.Plugin.Inventory.AllocationGate**
pessimistic row lock · reservations · expiry task
- **Nop.Plugin.Integration.Pos**
HTTP reserve / confirm / release
- **Nop.Plugin.Messaging.RabbitMq**
outbox pattern (OutboxMessage) + RabbitMQ publisher
- **Nop.Plugin.Fulfillment.OpenBoxes**
status poller → writes `carrier.booking.requested`
- **Nop.Plugin.Shipping.CarrierTracking**
shipment + carrier consumers · status poller · WireMock

Architectural Decisions

ESTABLISHED • STABLE (9)

ADR	Decision	Driver
001	RabbitMQ as message broker	QAS-1 Reliability
002	Plugin architecture as integration boundary	QAS-1 Reliability
003	Durable queues + persistent delivery + manual ack	QAS-1, QAS-4 Recoverability
004	Transactional outbox for reliable publish	QAS-1
005	Allocation Gate hosted inside nopCommerce	QAS-2 Consistency
006	Cross-channel allocation via synchronous HTTP	QAS-2 Consistency
007	OpenBoxes Bridge as separate deployable	QAS-1, QAS-4 Recoverability
008	Carrier status via scheduled polling	QAS-5 Visibility
009	OpenBoxes fulfillment state via scheduled polling	QAS-5 Visibility

● 9 stable

● 1 superseded

● 1 amended

● 1 new

CHANGED OR ADDED (3)

ADR 010

SUPERSEDED

Redis cache + distributed lock designed but **not implemented** — pollers read DB on every tick. Acceptable at single-node demo scale; required before horizontal scale-out.

ADR 007

AMENDED

Bridge consumer lifecycle changed from passive to active — **BridgeWorker** now calls **BasicConsumeAsync** / **BasicCancelAsync** based on circuit state.

ADR 011

NEW

Circuit breaker for the OpenBoxes Bridge — keeps transient failures in the durable main queue and self-heals via half-open probe, correcting a prior implementation that violated **QAS-4** by requiring manual DLQ replay.

Test Scenarios

Scenario	Pressure Applied	QAS Targeted
TS-1 Bridge Outage & Recovery	Bridge container stopped while 3 orders placed; restarted after outage	QAS-1, QAS-3, QAS-4
TS-2 Cross-Channel Oversell	POS reserves last unit via API while web customer attempts same product	QAS-2
TS-3 Idempotent Redelivery	Already-processed OrderGuid re-published directly to live RabbitMQ queue	QAS-1
TS-4 Carrier Status Propagation	WireMock state machine advances on each poll (IN_TRANSIT → OUT_FOR_DELIVERY → DELIVERED)	QAS-5
TS-5 Warehouse State Propagation	OpenBoxes fulfillment manually advanced to ISSUED	QAS-5
TS-6 Happy Path Baseline	No stress -> normal single order, all systems healthy	All QAS

Measurements

ID	QAS	Key Metric
M1	QAS-1 + QAS-4	0 orders lost; all processed within ≤ 60 s after recovery
M2	QAS-3	Checkout response time ≤ 3 s with OpenBoxes unreachable
M3	QAS-2	0 oversell events; exactly 1 winner
M4	QAS-5 carrier	Status change reflected in nopCommerce within ≤ 30 s
M5	QAS-5 warehouse	OpenBoxes ISSUED reflected in nopCommerce within ≤ 30 s
M6	QAS-1	0 duplicate fulfillments created in OpenBoxes

Experiment Results – Mandatory Pressure Point

Mandatory pressure point (M1):

- Bridge container stopped → 3 orders placed → bridge restarted
- Simulates OpenBoxes becoming fully unreachable during live traffic

Results:

- 0 orders lost: messages persisted on RabbitMQ (Ready=3, Consumers=0)
- All 3 fulfillments created in OpenBoxes within 60s of bridge restart
- 0 operator actions: bridge self-healed automatically
- DLQ remained empty throughout

Experiment Results – Mandatory Pressure Point

 RabbitMQ™
RabbitMQ 3.13.7 Erlang 26.2.5.16

Refreshed 2026-05-31 11:56:01 Refresh every 5 seconds Virtual host: All
Cluster rabbit@nopcommerce_rabbitmq
User guest Log out

[Overview](#) [Connections](#) [Channels](#) [Exchanges](#) [Queues and Streams](#) [Admin](#)

Queues

▼ All queues (4)

Pagination

Page 1 of 1 - Filter: ☐ Regex

Displaying 4 items, page size up to: 100

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	verdemart.carrier.booking.dlq	classic	D	running	0	0	0				
/	verdemart.carrier.booking.requested	classic	D DLX	running	0	0	0				
/	verdemart.orders.openboxes	classic	D DLX	running	3	0	3	0.00/s	0.00/s	0.00/s	
/	verdemart.orders.openboxes.dlq	classic	D	running	0	0	0				

▼ Add a new queue

Virtual host:

Type:

Name:

Durability:

Arguments: =

Add [Auto expire](#) [Message TTL](#) [Overflow behaviour](#) [Single active consumer](#) [Dead letter exchange](#) [Dead letter routing key](#) [Max length](#) [Max length bytes](#) [Leader locator](#)

[HTTP API](#) [Documentation](#) [Tutorials](#) [New releases](#) [Commercial edition](#) [Commercial support](#) [Discussions](#) [Discord](#) [Plugins](#) [GitHub](#)

Experiment Results – Mandatory Pressure Point

```
davidpelicano@davidpelicano-IdeaPad-Gaming-3-15ACH6:~/Desktop/nopCommerce$ docker stop verdemart_openboxes_bridge
verdemart_openboxes_bridge
davidpelicano@davidpelicano-IdeaPad-Gaming-3-15ACH6:~/Desktop/nopCommerce$ docker start verdemart_openboxes_bridge
verdemart_openboxes_bridge
davidpelicano@davidpelicano-IdeaPad-Gaming-3-15ACH6:~/Desktop/nopCommerce$ docker logs verdemart_openboxes_bridge 2>&1 | grep "Fulfillment created"
Fulfillment created for OrderGuid=84d42271-20f2-4c7c-bbe0-d53cfdad71ed FulfillmentId=2c9280869e7aa430019e7aa6b1a60002
Fulfillment created for OrderGuid=37ed6371-472d-4d38-984c-9b2410b1b0c7 FulfillmentId=2c9280869e7aa430019e7aadccf20005
Fulfillment created for OrderGuid=46902924-dce1-4778-97a9-8befadf77da9 FulfillmentId=2c9280869e7aa430019e7aadcd7b0009
```

Navigation: Main Warehouse | Dashboard | Inventory | Purchasing | Inbound | **Outbound** | Reporting | Products | Stocklists

Search: [Search icon] [Filter icon] [Settings icon] [User icon]

Outbound Movement List

[My Stock Movements](#) [Create Stock Movement](#)

Search by requisition number, name etc. [Hide Filters]

Clear [Search]

Requisition Status: [v] Main Warehouse: [v] Destination: [v] Shipment type: [v] Requested By: [v] Created By: [v] Updated By: [v] Request type: [v] Created after: [v] Created before: [v]

Outbound (9)

[Export](#)

# Items	Status	Identifier	Name	Destination	Stocklist	Requested By	Date Requested	Date Created	
...	0	Created	550XYR	Request > 8114a7a9-a92e-44de-9...	VerdeMart Store	None	Miss Administrator	May 31, 2026	May 31, 2026
...	0	Created	498BJF	Request > 1e51c5bc-9e9a-48a9-b...	VerdeMart Store	None	Miss Administrator	May 31, 2026	May 31, 2026
...	0	Created	266AFS	Request > 81f55419-7abf-49b3-98...	VerdeMart Store	None	Miss Administrator	May 31, 2026	May 31, 2026
...	0	Created	645VNC	Request > b505769e-be9c-4d25-9...	VerdeMart Store	None	Miss Administrator	May 30, 2026	May 30, 2026
...	0	Created	985FXB	Request > 37ed6371-472d-4d38-9...	VerdeMart Store	None	Miss Administrator	May 30, 2026	May 30, 2026
...	0	Created	121KCK	Request > 46902924-dce1-4778-9...	VerdeMart Store	None	Miss Administrator	May 30, 2026	May 30, 2026
...	0	Created	526YZX	Request > 84d42271-20f2-4c7c-b...	VerdeMart Store	None	Miss Administrator	May 30, 2026	May 30, 2026

Experiment Results – Checkout Availability + Oversell Prevention

Checkout availability (M2):

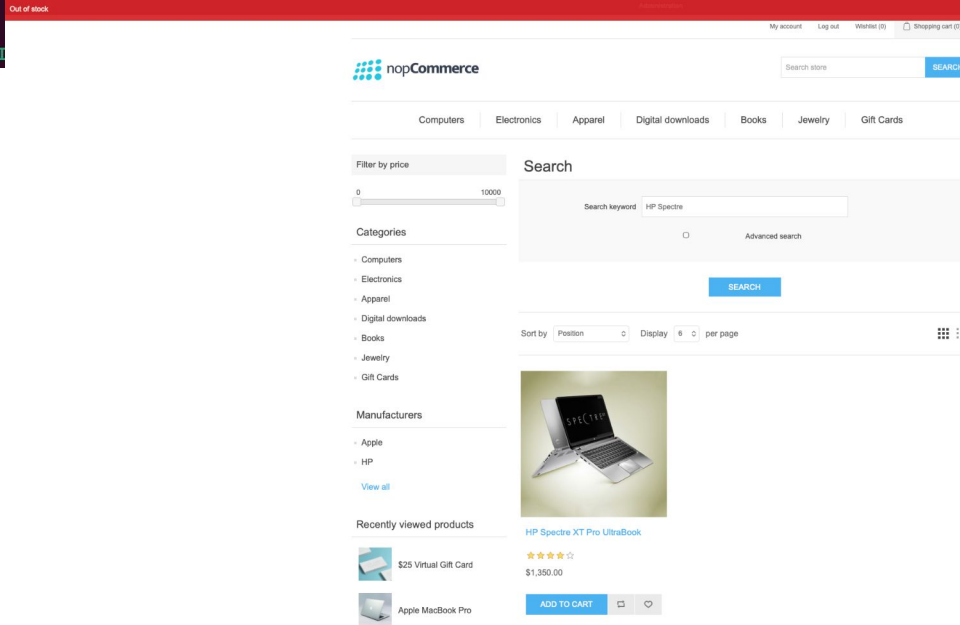
- Bridge stopped, OpenBoxes unreachable
- Checkout response time: 83–92 ms → 30× below the 3s QAS threshold

Cross-channel oversell prevention (M3):

- POS reserved the last unit → web channel blocked at product page ("Out of stock")
- StockQuantity never went below 1 in either scenario
- Rejection happened synchronously, within the same request cycle

Experiment Results – Checkout Availability + Oversell Prevention

```
davidpelicano@davidpelicano-IdeaPad-Gaming-3-15ACH6:~/Desktop/nopCommerce$ curl -s -X POST http://localhost/api/inventory/reserve -H "X-API-Key: verdemart-pos-key" -H "Content-Type: application/json" -d '{"productId": 7, "warehouseId": 0, "quantity": 1, "reservationKey": "m3-pos-1", "ttlSeconds": 300}' | python3 -m json.tool
{
  "reservationKey": "m3-pos-1",
  "message": "reserved"
}
davidpelicano@davidpelicano-IdeaPad-Gaming-3-15ACH6:~/Desktop/nopCommerce$ curl -s -X POST http://localhost/api/inventory/release \
-H "X-API-Key: verdemart-pos-key" \
-H "Content-Type: application/json" \
-d '{"reservationKey": "m3-pos-1"}' \
{"released":true}
```



Experiment Results – Visibility & Idempotency

Carrier status propagation (M4):

- WireMock state machine: Started → IN_TRANSIT → OUT_FOR_DELIVERY → DELIVERED
- All 3 transitions detected and written to nopCommerce within 30s
- Customer notification email queued atomically with each status update

Warehouse fulfillment visibility (M5):

- OpenBoxes stock movement advanced to ISSUED manually
- Order marked Complete in ~18s: Shipment row created automatically
- Full cascade triggered with no operator action: ISSUED → Complete → carrier booked → DELIVERED

Idempotent redelivery (M6):

- Duplicate message re-published to live queue
- Bridge: "already processed — ack and skip": OpenBoxes API never called
- Zero duplicate stock movements created

Experiment Results – Visibility & Idempotency

RabbitMQ 3.13.7 Erlang 26.2.5.16

Overview Connections Channels Exchanges **Queues and Streams** Admin

Effective policy definition

Consumers (1)

Channel ?	Consumer tag	Ack required	Exclusive	Prefetch count	Active ?	Activity status	Consumer Timeout	Arguments
172.18.0.8:47448 (1)	amq.ctag-4HhL5nRFiUHW3F_vdMLAA	☒	☐	10	☐			

Bindings (2)

Publish message

Message will be published to the default exchange with routing key `verdemart.orders.openboxes`, routing it to this queue.

Delivery mode: **2 - Persistent**

Headers: ? = String

Properties: ? =

Payload:

```
{
  "OrderId": 1,
  "OrderGuid": "46743bbe-e60f-497e-8cc7-ec801481c72d",
  "CustomerId": 1,
  "OrderTotal": 100.00,
  "CreatedOnUtc": "2026-05-01T00:00:00Z",
  "Items": [
    {
      "ProductId": 4,
      "Sku": "AP_MBP_13",
      "Name": "Apple MacBook Pro",
      "Quantity": 1,
      "UnitPriceInclTax": 100.00
    }
  ],
  "Version": 2
}
```

Payload encoding: String (default)

Publish message

OrderGuid=f69aee6c-adbe-4f1a-a3c7-7490ce5e5a93 already processed – ack and skip
OrderGuid=46743bbe-e60f-497e-8cc7-ec801481c72d already processed – ack and skip

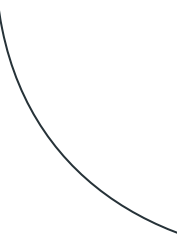
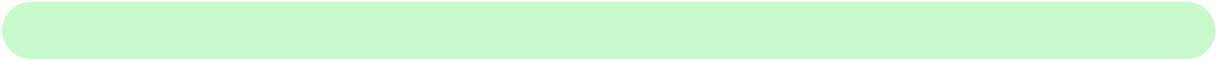
Known Limitations

Accepted trade-offs

- Redis distributed lock not implemented: pollers read DB on every tick; unsafe at multi-node scale (ADR-010 specified but not built)
- No shared HTTP policy: each plugin manages its own HttpClient; no circuit breaker across integrations
- Stock back-propagation OpenBoxes → nopCommerce not implemented: warehouse corrections do not update nopCommerce stock quantities

Discovered during testing

- SQL deadlocks under ≥ 5 concurrent POS requests on the same product row — zero oversell preserved, but error degrades from structured 409 to HTTP 500



DEM
O

Conclusion

Architectural evolution of nopCommerce from a single-channel storefront into a resilient omnichannel commerce core

Five quality attribute scenarios drove all design decisions: each with a traceable mechanism and empirical validation

Mandatory pressure point exercised in full: degradation contained, no data loss, autonomous recovery demonstrated

Selective scope: focused architectural changes with clear justification for what remains inside the monolith

Architectural Evolution of nopCommerce

David Pelicano ◉ Henrique Ferreira ◉ João Monteiro ◉ Jorge Domingues

05.05.2026

